Diseases of the Male Reproductive System

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Prostate and testicular disorders overview → Prostatitis (acute vs. chronic) → Benign prostatic hyperplasia (BPH) → Comparing the two

This lecture focuses on two prostate disorders: benign prostatic hyperplasia (BPH) and prostatitis. The prostate's location explains why both conditions affect urination. Connect changes in the prostate to urinary flow, bladder emptying, and the symptoms that result.

A note on language: The slides use "male" and "female" to describe reproductive anatomy. Anatomy does not tell us someone's gender identity, and not everyone has all the structures shown. There is a spectrum of what is normal in anatomy. In practice, use the person's name, pronouns, and preferred terms, and focus on the anatomy relevant to their care.

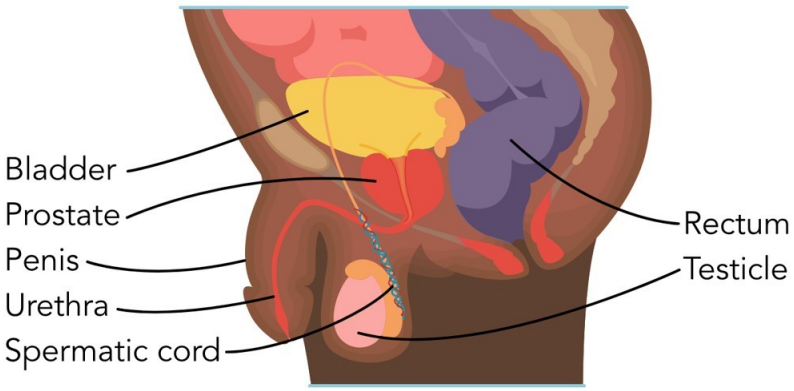
Learning Objectives (from the slides)

- Sexually transmitted infections: etiology, pathogenesis and pathophysiology, manifestations, and diagnostic criteria.
- Prostate disorders: the same elements.
- Testicular disorders: the same elements.

Heads up: This lecture only covers prostatitis and BPH. The ATI Engage module also lists prostate cancer and testicular disorders. Prostate cancer is covered later in the semester. Sexually transmitted infections and testicular disorders are not covered in these slides or in the lecture; use your ATI module for those.

Prostate and Testicular Disorders: Overview

The ATI Engage module lists prostatitis, BPH, prostate cancer, and testicular disorders. Prostate disorders include both benign and malignant conditions; BPH and prostatitis are benign in the sense that they are typically not associated with cancer.



Male reproductive and urinary anatomy: bladder, prostate, penis, urethra, spermatic cord, testicle, rectum (lecture slide)

Why Location Matters

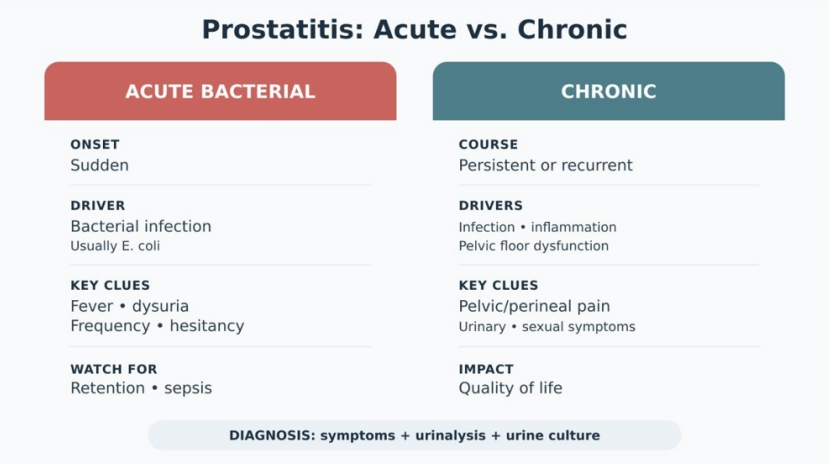
- **Position:** the prostate sits just below the bladder.
- **Urethra:** passes through the prostate before continuing toward the outside of the body at the urethral meatus.
- **Result:** enlargement or swelling of the prostate can increase resistance to urine flow.

Condition	What happens in the prostate
BPH	An increase in prostate tissue
Prostatitis	Inflammation, and it may involve a bacterial infection

Both can affect urination, but they develop through different mechanisms.

Prostatitis

Conditions involving inflammation of the prostate, or symptoms such as pelvic pain and urinary difficulty. Divided into acute and chronic. Acute is short-term with rapid onset; chronic is persistent.



Prostatitis: acute vs. chronic (lecture slide)

	Acute bacterial	Chronic
Onset and course	Sudden, rapid onset	Persistent or recurrent
Cause	Bacteria enter the prostate tissue, often from the urinary tract. E. coli is a common cause. Bacteria can also be introduced during urinary procedures such as catheterization	Usually persistent or recurrent bacterial infection, but it does not always involve bacteria. Can come from inflammation with other causes or pelvic floor dysfunction (chronic pelvic pain syndrome)
Key findings	Fever, chills, feeling generally ill; dysuria and urinary frequency; pelvic or perineal pain; hesitancy, difficulty urinating, or urinary retention	Symptoms may be less dramatic. Pelvic or perineal pain, urinary symptoms, recurrent UTIs (a clue), sexual symptoms such as pain with ejaculation
Watch for	Retention, sepsis	Impact on quality of life
Diagnosis	Primarily the presence of bacteria in the urine (urinalysis and urine culture)	Mostly client-reported manifestations and how long they have been going on. Urinalysis and culture help evaluate for bacterial infection

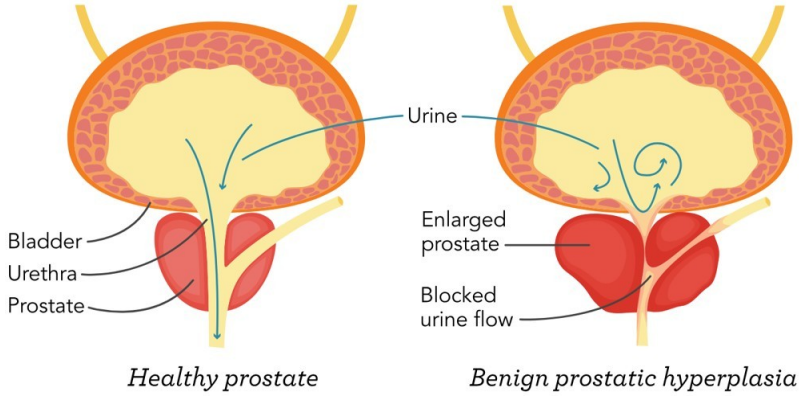
Pathophysiology: Connect the Dots

1. **Trigger:** infection or another cause produces inflammation and swelling of the prostate.
2. **Pain:** inflammation explains pelvic or perineal pain.
3. **Obstruction:** swelling around the urethra changes urinary flow (hesitancy, difficulty urinating, even retention).
4. **Systemic signs:** in acute infection, fever, chills, and malaise.

Exam note: Acute prostatitis is very likely caused by a bacterial infection. Chronic prostatitis (or chronic pelvic pain syndrome) does not always involve bacteria, and sexual side effects (pain with ejaculation) and pelvic floor pain are more likely on the chronic side, which helps tell the two apart.

Benign Prostatic Hyperplasia (BPH)

Hyperplasia means an increase in the number of cells. This is not swelling from inflammation, fluid, and cellular products as in prostatitis. In BPH there is an actual increase in prostate tissue.



Healthy prostate vs. BPH: enlarged prostate blocks urine flow (lecture slide)

Etiology

- **Age:** becomes more common with aging; very common in men (people with a prostate) over 60.
- **Hormones:** hormonal influences on tissue growth.
- **Other risk factors:** diabetes and associated medication therapies, a history of chronic prostatitis, and obesity.

Pathogenesis and Pathophysiology

1. **Tissue growth:** a discrepancy between cellular growth and cellular apoptosis (cell death) leaves more tissue.
2. **Location:** the enlargement typically develops around the part of the urethra passing through the prostate.
3. **Narrowing:** growing tissue narrows the passage and increases resistance to urine flow. Smooth muscle tone in the prostate also adds to the resistance.
4. **Bladder response:** the bladder must generate more pressure to push urine through. Over time the bladder muscle may thicken and bladder function can change.

Manifestations

Manifestation	Why
Hesitancy, weak or interrupted stream, straining, feeling of incomplete emptying	Difficulty emptying because of the narrowed urethra
Urgency, frequency, nocturia (getting up at night to urinate)	Bladder changes

Size does not equal severity: Prostate size and symptom severity do not always match. The location of the enlargement, the degree of muscle tone change, and the bladder's response all play a role.

Diagnostic Criteria and Assessment

- **Pattern of urinary symptoms:** start here. Also consider other causes such as infection, medications, or other bladder dysfunction.
- **Urinalysis:** identifies blood or evidence of infection.

- **Prostate exam:** provides information about the gland.
- **Post-void residual:** measures how much urine remains in the bladder after urinating.

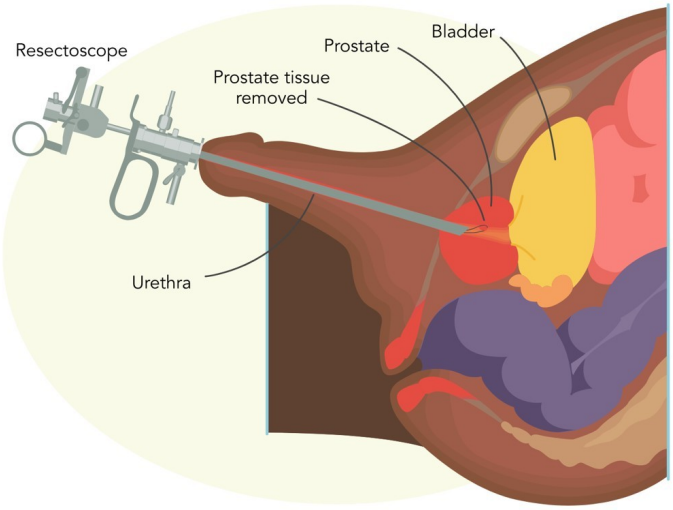
Impact on the Health of the Client

- **Retention and stasis:** persistent retention can contribute to urinary stasis, which can lead to infection. The instructor also said it can increase kidney stones.
- **Back pressure:** more severe obstruction creates back pressure that affects the upper urinary tract and the kidneys.
- **Daily life:** frequent night-time urination disrupts sleep, and symptoms can disrupt work activities.

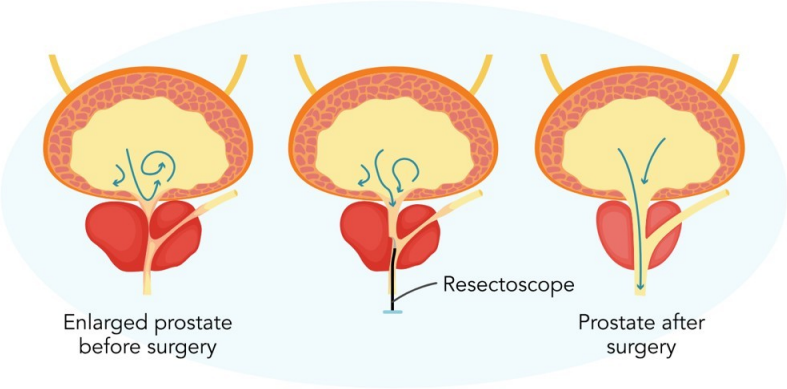
Focus for this course: Understand what BPH is, the changes you will see, and the possible complications.

Management (awareness only)

Management was not discussed. The slides show a resectoscope inserted through the urethra to remove prostate tissue, which restores urine flow.



Resectoscope removing prostate tissue through the urethra (lecture slide)



Enlarged prostate before surgery, during, and after: urine flow restored (lecture slide)

Wrapping Up: BPH vs. Prostatitis

	BPH	Prostatitis
Underlying process	Increase in tissue (cell growth exceeds apoptosis), plus smooth muscle effects	Inflammation, sometimes from bacterial infection
How it affects urination	Increased resistance to urine flow	Swelling around the urethra changes urinary flow
Main complaint	Hesitancy, weak stream, nocturia, incomplete emptying	Pelvic or perineal pain, urinary symptoms, fever and chills if acute
Typical risk	Older (over 60), hormonal influence	Acute: E. coli or urinary procedures such as catheterization

Tie it together: Both may interfere with bladder emptying. Connect incomplete emptying to urinary stasis, infection risk, and (in severe obstruction) effects on the kidney. Link this back to the other conditions covered this week: obstruction, stasis, UTIs, kidney stones, and back pressure on the kidneys.

Quick Review

Condition	Key points
Prostatitis, acute	Sudden; bacterial (E. coli, catheterization/procedures); fever, chills, malaise, dysuria, frequency, pelvic/perineal pain, hesitancy, retention; diagnosed by bacteria in urine (UA and culture); watch for retention and sepsis
Prostatitis, chronic	Persistent or recurrent; infection, inflammation, or pelvic floor dysfunction (not always bacterial); pelvic pain, recurrent UTIs, pain with ejaculation; diagnosed by symptoms and duration
BPH	Increase in prostate tissue (hyperplasia); aging (over 60), hormones, diabetes, obesity, chronic prostatitis; narrows urethra; hesitancy, weak interrupted stream, straining, incomplete emptying, urgency, frequency, nocturia; size does not equal severity; PVR, UA, prostate exam; complications: stasis, infection, stones, kidney back pressure

Notes on Transcript Cleanup

"Chronic prostatitis can also lead to BPH and obesity" was written as chronic prostatitis and obesity being additional BPH risk factors. The lecture speaker is not named on the title slide, so the byline follows the other renal and urinary lectures. Management, sexually transmitted infections, and testicular disorders were not covered.